



Silicon Info

A CPU, GPU and Neural Engine widget for Apple Silicon Macs

Documentation, version 1.1

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1. Overview

Silicon Info breaks Mac activity into the pieces that make up an Apple Silicon chip instead of showing one combined number. It comes in three forms that share the same data: a floating panel on the desktop, a real desktop widget, and a menu bar item with the live CPU load. The panel has five cards:

- **CPU** performance cores and efficiency cores, shown separately.
- **GPU** device utilization with render and tiler load.
- **Neural Engine** power draw and activity history.
- **Memory** unified memory (DRAM) usage, pressure and swap.
- **Power** how total power divides between the three blocks.

Each card is a summary. Click a card to expand it into a detailed view, and use the Back control to return. The widget is available in three sizes.

2. Requirements

Item	Requirement
Hardware	Mac with Apple Silicon (developed on an M4 Max)
Operating system	macOS 14 or later
Build tools	Xcode and XcodeGen (<code>brew install xcodegen</code>)
Code signing	An Apple Developer team. Widgets and App Groups do not work with ad hoc signing.
Privileges	An administrator password, entered once per launch, for power and frequency data and for freeing cached memory

3. Installation and use

Build

Set your own team in `project.yml` (`DEVELOPMENT_TEAM`), and change the bundle identifiers and the App Group (`875N49PYZ9.com.raymondmunro.siliconinfo` , in `project.yml` and `Sources/Shared/WidgetData.swift`) to match your team. Then:

```
./build.sh
```

This generates the Xcode project from `project.yml` , then builds and signs `SiliconInfo.app` together with its widget extension.

Run

```
open build/Build/Products/Release/SiliconInfo.app
```

On launch the app asks for your administrator password through the standard macOS dialog. The floating panel appears at the top right of the main display. The app has no Dock icon, but it adds a menu bar item. Drag the panel anywhere by its background. Its position is remembered, and the window is kept fully on screen when a card expands.

Hide process names

The CPU, GPU and Memory views list process names. To hide those lists, for example before sharing your screen, start the app with the privacy option:

```
build/Build/Products/Release/SiliconInfo.app/Contents/MacOS/SiliconInfo --hide-processes
```

Quit

Use Quit Silicon Info in the menu bar menu, or run:

```
pkill SiliconInfo
```

4. The views

Overview

Card	Shows
CPU	One bar per core, grouped into performance cores (blue) and efficiency cores (green), each group's average load, package power and a load history line.
GPU	A ring for device utilization, render and tiler bars, power and a history line.
Neural Engine	A ring for power draw and a history line.
Memory	Used and total memory, a bar split into app, wired and compressed memory, and the current pressure level.
Power	A single bar divided between CPU, GPU and Neural Engine, with the total in watts.

CPU detail

Section	Contents
Clusters	For the efficiency cluster and each performance cluster: current frequency, active share, and a histogram of time spent at each clock step.
Cores	Every core with a stacked bar of user time (blue or green) and system time (red), its current clock speed and total load.
Top processes	The busiest processes, as a percentage of one core.
Power	Two minutes of CPU power with average and peak.

GPU detail

Section	Contents
Overview	Utilization ring, current frequency, maximum clock, active and idle share, power.
Frequency residency	Time spent at each GPU clock step.
Engines	Device, render and tiler utilization.
Memory	GPU memory in use and allocated. This memory is shared with the system.
Top GPU processes	Processes using GPU time, as a share of the GPU.

Neural Engine detail

Shows current power, average and peak over two minutes, energy used since launch, an active or idle state and a power history chart. Load is power as a share of an assumed 8 W ceiling, so treat the ring as relative.

Memory detail

Apple Silicon uses one pool of unified memory shared by the CPU, GPU and Neural Engine, so this is the DRAM view. Viewing it needs no administrator rights. The Free memory button does.

Section	Contents
Unified memory	Used, available and total memory, with a bar and a legend for app memory, wired, compressed, cached files and free. Used memory follows the Activity Monitor definition: app plus wired plus compressed.
Free memory	A button that frees cached memory, with the amount freed shown beside it. See the Free memory section below.
Pressure and swap	The system pressure level (normal, warning or critical), swap in use and swap size, and the rate of data paged in and out.
Top processes	The processes holding the most resident memory, refreshed every five seconds.
History	Two minutes of memory use.

Power detail

Shows the live split between CPU, GPU and Neural Engine, a stacked two-minute history where the top edge is the total, and average power, peak power and energy in milliwatt-hours for each block since launch.

5. Menu bar and desktop widget

Menu bar

The app adds a menu bar item that shows the live total CPU load next to its icon. Its menu shows live CPU, GPU, Neural Engine and memory readings, Free Cached Memory, a Hide Panel or Show Panel toggle for the floating panel, and Quit Silicon Info. If you use a menu bar manager, the icon may be inside its hidden section.

To get rid of the floating panel, right-click it and choose Hide Panel, or use the menu bar item. It stays hidden across launches, and the desktop widget and menu bar item keep working. Opening the app again from Finder or Spotlight shows the panel.

Desktop widget

Silicon Info includes a WidgetKit widget. Run the app once, then right-click the desktop, choose Edit Widgets and search for Silicon Info.

Size	Shows
Small	One component of your choice: CPU, GPU, Neural Engine, Memory or Power. Edit the widget to pick which. When it shows Memory, tapping it frees cached memory.
Medium	CPU cores, GPU and Neural Engine side by side, with memory and a power split. Tap the memory bar to free cached memory.
Large	CPU cores, GPU, Neural Engine, memory and power. Tap the memory panel to free cached memory.

Free memory

Tapping the memory area of the widget frees cached memory. The same action is available as a button in the Memory detail view and as Free Cached Memory in the menu bar menu. The widget then shows how much was freed for about a minute.

This runs `purge`, which empties the disk cache (the Cached files slice). It does not free memory used by apps, since only quitting apps does that, and the cache refills as you work, so things may feel briefly slower. It needs the administrator prompt to have been approved at launch, otherwise the widget says so.

How it works

A widget runs in a sandbox and cannot read hardware counters or run `powermetrics`. The Silicon Info app therefore keeps sampling and writes a small snapshot file into an App Group container that both the app and the widget are entitled to use. The widget reads that file whenever macOS redraws it.

macOS draws desktop widgets in grey when another app is in front. To keep them in colour, set Widget style to Full-color in System Settings, under Desktop & Dock.

macOS limits how often widgets redraw, so expect an update every several seconds, not every second like the floating panel. Keep the app running for fresh data. If it is not running, the widget shows a small note asking you to open it.

6. Data sources

Data	Source	Needs admin
Per-core load, user and system split	Mach <code>host_processor_info</code>	No
GPU utilization, render, tiler, memory	IOKit accelerator statistics	No
Memory breakdown, pressure, swap, paging rates	Mach <code>host_statistics64</code> and <code>sysctl</code>	No
Top processes by memory	<code>ps</code>	No
CPU, GPU and Neural Engine power	<code>powermetrics</code>	Yes
Core and cluster frequency, residency	<code>powermetrics</code>	Yes
GPU frequency and residency	<code>powermetrics</code>	Yes
Per-process CPU and GPU time	<code>powermetrics</code> tasks sampler	Yes
Freeing cached memory	<code>purge</code>	Yes

Core order follows Apple Silicon convention: efficiency cores are numbered first, then performance cores.

7. Privileges and safety

Reading power and frequency needs root access. On launch the app shows the standard macOS administrator dialog. After approval, a small shell loop runs as root. It does two things: it runs `powermetrics` once per second and writes the text output to `/private/tmp/siliconwidget-powermetrics.txt`, which the app reads, and it runs `purge` when asked to free cached memory.

- The loop exits on its own about six seconds after the last Silicon Info process quits.
- Relaunching quickly reuses the running loop and does not ask again.
- Nothing is sent over the network. The output file contains only power, frequency and process names with CPU and GPU time.
- If you cancel the dialog, the app keeps running and shows `n/a` for the values that need it. Free Cached Memory then reports that it needs admin access.
- The privileged loop also runs `/usr/sbin/purge` when the app creates a trigger file whose name contains a secret generated for your install. Only the app knows the name, so other local users cannot trigger it.

The output file is readable by other local users, so process names on the machine are visible to them while the app runs.

8. Limitations

macOS does not expose the following on this hardware, so they are not shown:

- **Neural Engine utilization, frequency or per-process use.** Only power draw is available.
- **Per-cluster CPU power.** Only the CPU total is reported.
- **DRAM bandwidth and DRAM power.** macOS refuses the counter subscription even with administrator rights, so only memory usage is shown.
- **Media engine power.** It reads zero on this chip.

Per-process GPU time is often zero because few processes keep the GPU busy at any moment.

9. Troubleshooting

Symptom	Cause and fix
Power shows <code>n/a</code>	The administrator dialog was cancelled. Quit and relaunch, then approve it.
Power graphs are empty after launch	History fills at one sample per second. Wait a minute.
Neural Engine reads 0 W	The engine is idle. Run something that uses it, such as Core ML work, to see it move.
Floating panel is off screen	Delete the saved position with <code>defaults delete com.raymondmunro.siliconinfo</code> and relaunch.
Widget says "Open Silicon Info to update"	The app is not running, so no fresh snapshot is being shared. Start the app.
Free Cached Memory says "Needs admin access"	The administrator dialog was cancelled or the privileged loop has stopped. Quit and relaunch, then approve the dialog.
Desktop widget looks grey	macOS draws widgets in grey when another app is in front. Set Widget style to Full-color in System Settings, under Desktop & Dock.
Floating panel is gone	It was hidden. Use Show Panel in the menu bar menu, or open the app again from Finder or Spotlight.
Widget is missing from the gallery	Run the app once. If it still does not appear, copy the app to the Applications folder and open it from there.
Build fails	Install the command line tools with <code>xcode-select --install</code> .

10. Project layout

Path	Purpose
<code>Sources/SiliconInfo/main.swift</code>	Creates the borderless floating window and the menu bar item, and keeps the window anchored and on screen.
<code>Sources/SiliconInfo/Sampler.swift</code>	Samples core load and GPU statistics each second and holds the history.
<code>Sources/SiliconInfo/Memory.swift</code>	Reads unified memory usage, pressure, swap and the top processes by memory.
<code>Sources/SiliconInfo/MemoryPurger.swift</code>	Hands free memory requests to the privileged loop and reports how much was freed.
<code>Sources/SiliconInfo/PowerFeed.swift</code>	Starts the privileged loop and parses the <code>powermetrics</code> output.
<code>Sources/SiliconInfo/Views.swift</code>	The overview cards and expand and collapse behavior.
<code>Sources/SiliconInfo/Detail.swift</code>	The CPU, GPU, Neural Engine, Memory and Power detail views.
<code>Sources/SiliconInfo/WidgetBridge.swift</code>	Builds the snapshot shared with the widget and asks WidgetKit to redraw.
<code>Sources/SiliconInfoWidget/</code>	The WidgetKit extension: timeline provider, widget configuration and the tap to free memory action.
<code>Sources/Shared/</code>	Data model, shared drawing code and the small, medium and large layouts, used by both targets.
<code>project.yml</code>	Xcode project definition, generated by XcodeGen.
<code>scripts/make_icon.swift</code>	Draws the app icon.
<code>build.sh</code>	Generates the project and builds the signed app with its widget.

11. License

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